

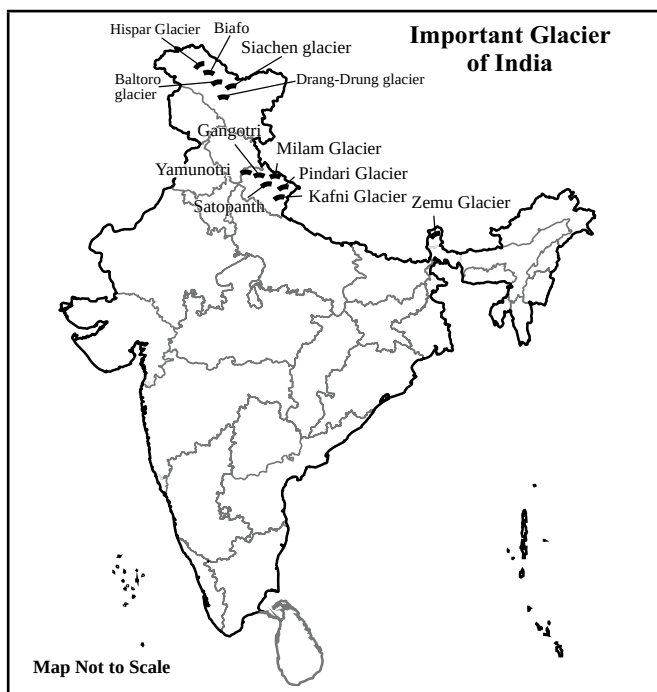
The correct match is -

<u>Pass</u>	<u>Height from sea levels</u>	<u>States/UTs</u>
Debsa	5360 meters	Himachal Pradesh
Rohtang	3978 meters	Himachal Pradesh
Bara-Lacha	4850 (Approx 4890 meters)	Himachal Pradesh
Banihal	2832 meters	Jammu and Kashmir

vii. Snow Lines and Glaciers

*The altitude at a particular place above which some snow remains on the ground throughout the year is called a snowline. A glacier is a slowly moving mass or river of ice formed by the accumulation and compaction of snow on mountains or near the poles.

*Different areas of the Himalayas have different snowlines depending upon the differences in latitude, relief, precipitation, slope and local topography. However, the average snowline is found at a height of **5500-6000** meters in the northern part of the Greater Himalayas and between **4500 - 6000** meters in the southern part of Greater Himalayas. Approximately 40,000 sq. km area of the middle Himalayas ranging from Karakoram in Ladakh to Arunachal Pradesh is covered with glaciers.



*The most important glaciers of Himalayas are located in the mountains of Great Himalayas and Trans Himalayas. Glaciers are found in the Karakoram, Ladakh and Zaskar

ranges. *The glaciers of lesser Himalayas are small in size, however, the Pir Panjal and Dhauladhar ranges provide evidence of larger glaciers. The largest glacier of Pir Panjal range is **Sonapani**, which is situated in the Chandra Ghati of Lahul and Spiti district in Himachal Pradesh. *The Karakoram and Greater Himalayas consist of immense glaciers, **Siachen** (76.64 km), **Hispar** (61 km), **Biafo** (60 km), **Baltoro** (58 km) and **Sasaini** (17.85 km) are some prominent ones. The **Chorabari glacier** is situated in the state of Uttarakhand in the district of Rudrapur in the Garhwal Himalaya region of the state of Uttarakhand, north of the famous temple of Kedarnath. The southern slope of this glacier is located to the north of the temple of Kedarnath at a distance of approx. 4 km. Due to melting of the Chorabari glacier, a glacial lake was formed, known by the name of Gandhi Sarovar. *The **Milam Glacier** is located in the Kumaon region of Uttarakhand. It is the main glacier of the area and the source of River Gori Ganga (Kali River).

Glacier - which is larger Siachen or Sasaini

* According to UPPSC - Siachen (source - Geography of India, Ramchandra Tiwari).

* According to Allahabad High Court - Sasaini

Source - Bhautik Bhugol Ka Swaroop - Author Ravindra Singh

- Geography of India by V.S. Chauhan & Alka Gautam

- Geography of India by K. Bharadwaj

Now the books are efficient enough as a source for determining the measurement of the glaciers while sources based on real surveys are available regarding the length and area of glaciers.

Sources such as -

1. Records of the Geological Survey of India (Vol. 63)
2. World Glacier Inventory.

❑ Decoding to the records of Geological Survey of India-

1. Siachen (L-45 miles)
2. Sasaini - (L-11 miles).

● According to the Geological Survey of India (Vol. 63, Pg. - 260) - outside the poles the largest glacier is **Fedchenko**, situated in the Pamir Region, whereas **Siachen** is the second largest glacier. Even according to the various sources available on the internet Fedchenko is the largest glacier outside the polar region and Siachen is the second largest. Hence, according to this order Sasaini is way below in the table.

❑ According to World Glacier Inventory -

- Siachen - L - 76.64 km, area - 1056.42 km²
- Sasaini - L - 17.85 km, area - 33.51 km².
- According to W.G.I. even Zemu and Gangotri glaciers are larger than Sasaini.

Hence the claim of Sasaini being a larger glacier than Siachen is debatable.

1. The snow-line in Himalayas lies between

- (a) 4300 to 6000 meters in East
- (b) 4000 to 5800 meters in West
- (c) 4500 to 6000 meters in West
- (d) None of the above

39th B.P.S.C. (Pre) 1994

Ans. (a)

The altitude in a particular place above which some snow remains on the ground throughout the year is called snow line. The snow line in the Himalayas has different heights in different parts. On an average it has height of 5500 – 6000m in Northern Part and 4500- 6000m in Southern Part of the Himalayas. In this way snow line in Himalayas lies between 4300 m to 6000 m.

2. Average height of snow-line in Uttar Pradesh is:

- (a) 6500 m
- (b) 7500 m
- (c) 5500 m
- (d) 4500 m

70th B.P.S.C Re-Exam 2024

Ans. (*)

The snow line refers to the lowest elevation at which permanent snow and ice typically occur in a region. There is no snowfall in Uttar Pradesh after the separation of Uttarakhand. None of the options are correct, so the question is deleted.

3. Siachen Glacier is situated to the:

- (a) East of Aksai Chin
- (b) East of Leh
- (c) North of Gilgit
- (d) North of Nubra Valley

I.A.S. (Pre) 2020

Ans. (d)

Siachen Glacier is located in the Eastern Karakoram range in the Himalayas, and it lies to the North of Nubra Valley. The Siachen glacier (highest battlefield in the world) demarcates central Asia from the Indian subcontinent, and separates Pakistan and China in the region.

4. Which of the following is the largest glacier ?

- (a) Siachen
- (b) Baltoro

(c) Chogo Lungma

(d) Biafo

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

Siachen is the largest glacier among the given options. It is spread over 1056.42 sq. km.

5. Which of the following is the largest glacier?

- (a) Kanchenjunga
- (b) Rundun
- (c) Gangotri
- (d) Kedarnath

U.P.P.C.S. (Pre) 1995

Ans. (c)

The lengths of the above glaciers are given below :

Gangotri	- 30.0 km
Rundun	- 19 km
Kanchenjunga	- 16 km
Kedarnath	- 14.5 km

In the above question, Gangotri is the largest/longest glacier. So option (c) is correct.

6. Which amongst the following is the largest glacier?

- (a) Sasaini
- (b) Gangotri
- (c) Zemu
- (d) Siachen

U.P.P.C.S. (Pre) 2017

Ans. (d)

Siachen is the largest glacier in India. According to the textbooks titled as 'Bhautik Bhugol ka Swaroop' written by Savindra Singh, 'Geography of India' written by V.S. Chauhan and Alka Gautam and 'Physical Geography (Introduction to Earth)' written by K. Bharadwaj and 'Bharat Ka Bhugol (Geography of India)' written by Professor Ram Chandra Tiwari, the length of Siachen is 72 km. According to the World Glacier Inventory, the length of Siachen is 76.64 km.

7. Chaurabari Glacier is located towards –

- (a) South of Kedarnath temple
- (b) West of Kedarnath temple
- (c) North of Kedarnath temple
- (d) East of Kedarnath temple

U.P.P.C.S. (Mains) 2004

U.P.P.S.C. (GIC) 2010

Ans. (c)

Chaurabari Glacier is located in Garhwal Himalayas region of Uttarakhand. The glacier lies in north of the Kedarnath Temple. Due to melting of glacier, a lake is formed which is named as Gandhi Sarovar Lake.